

EXPERIMENT NO: 8

User Management and Role-Based Access Control

Project: Student Admission Enquiry Form
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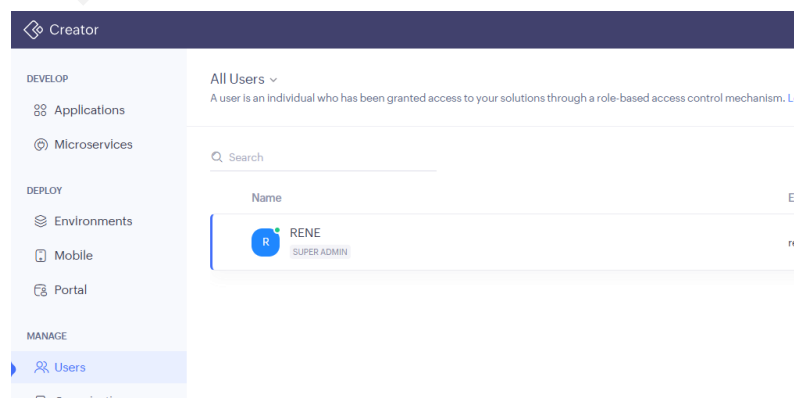
AIM

To implement user management and role-based access control (RBAC) in the Student Admission Enquiry Form application using Zoho Creator, by adding users, defining permission profiles, creating a role hierarchy, and configuring data sharing rules.

PROCEDURE

1. Open Zoho Creator and go to **Manage > Users** from the left navigation panel. The list of existing users is displayed. Only the Super Admin or an authorised administrator can open this section.
2. Click **Add User**. In the panel that opens, enter the **Email Address** and **Name** of the new user, and tick *Send Notification Mail* if an invitation is required.
3. Under **Add to Application** (tab *Applications as User*), select the **Application Name** (Student Admission Enquiry Form), choose the **Role** and choose the **Permission** for the user. Click **Add User** to save. The new user receives access to the application with exactly the rights of the chosen role and permission.
4. Open the application and go to **Settings > User Permissions > Permissions**. Here the permission profiles (a Read profile and the Write profile) are available; new profiles can be added with **Add Permission**. Every user in the application must be linked to one of these permission profiles.
5. Go to the **Roles** tab and build the role hierarchy (CEO > Admission Manager > Admission Staff) using **Add Role** and **Manage Role Hierarchy**. The hierarchy decides which users can see the records of other users.
6. Go to the **Data Sharing** tab, click **New Share Rule**, and specify the form, the role the data is shared from, the role it is shared to, and the level of access (Read/Write). Save the rule; it becomes active immediately for the selected role.

Step 1 – All Users list



The **All Users** page under **Manage > Users** lists every user who has been given access to the solutions through role-based access control. It shows each user's name and email address. **RENE** is the **Super Admin** of the account and manages all other users.

Step 2 & 3 – Adding a user to the application

The **Add User** panel in Zoho Creator collects the user's email address and name, and lets the administrator assign the application, the role and the permission under *Applications as User*. The left panel shows the Manage section with the existing user RENE (Super Admin).

Step 4 – Permission profiles

Under **Settings > User Permissions > Permissions**, the Student Admission Enquiry Form application has two profiles: **rene** (read permission for all components) and **Write** (write permission for all components). The **Add Permission** button creates custom profiles.

OBSERVATIONS ON THE PROCEDURE

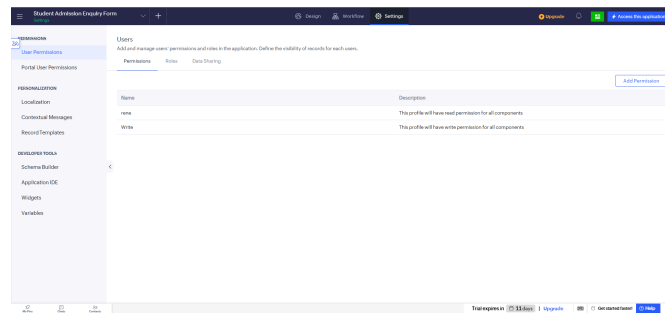
While adding a user, the email address uniquely identifies the person, and the notification mail informs them that they have been invited to the application. Without selecting an application, role and permission, the user exists in the account but cannot open the Student Admission Enquiry Form application. Once the role and permission are chosen, the user can log in and works only within the limits set by the administrator.

The All Users page gives a quick view of everyone who can access the account. The Super Admin can search for a user and check whether each user is active before assigning a role or permission. The Read and Write permission profiles separate people who only need to view admission enquiries from people who must create or edit them, and custom profiles can be created with Add Permission when finer control over individual forms, reports or pages is needed.

Permissions, roles and data sharing rules are configured separately, so a change in one of them does not disturb the others. This makes access control flexible: a user can be moved to a different role without redefining the permission profile, and a new share rule can be added without changing the role hierarchy.

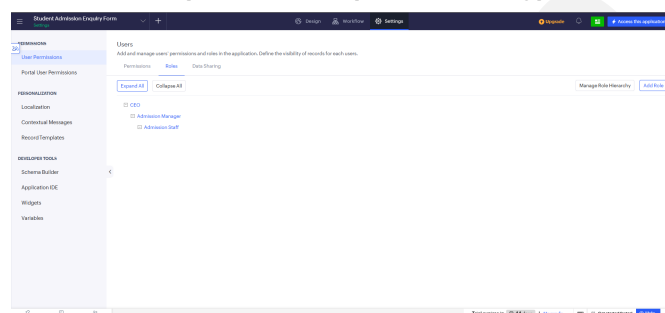
OUTPUT

Output 1 – Permissions



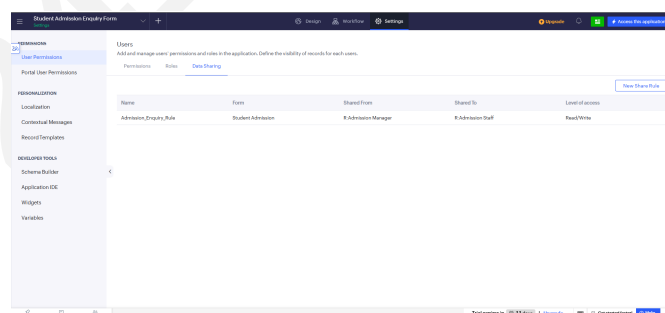
The **Permissions** tab lists the available permission profiles. The *rene* profile grants read access and *Write* grants write access to all components of the application. These profiles are assigned to users while adding them.

Output 2 – Roles (Role Hierarchy)



The **Roles** tab shows the role hierarchy of the application: **CEO** at the top, **Admission Manager** below the CEO, and **Admission Staff** at the lowest level. Users higher in the hierarchy can view the records of the roles beneath them. Roles can be added with *Add Role* and rearranged with *Manage Role Hierarchy*.

Output 3 – Data Sharing



The **Data Sharing** tab shows the share rule named **Admission Enquiry Rule** created for the form **Student Admission**. Records of the role **Admission Manager** (Shared From) are shared with the role **Admission Staff** (Shared To) with **Read/Write** access. This defines the visibility of records beyond the default role hierarchy.

RESULT

Thus, user management and role-based access control were successfully implemented in the Student Admission Enquiry Form application. Users were added with assigned roles and permissions, a role hierarchy (CEO, Admission Manager, Admission Staff) was created, and a data sharing rule was configured to control record visibility and access levels.

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